



Decentralized Evaluation of Regular Expressions for Capability Discovery in Peer-to-Peer Networks

Maximilian Szengel

Advisors: C. Grothoff, R. Holz, H. Niedermayer, B. Polot

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Chair for Network Architectures and Services
Technische Universität München

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Searching in DHT-based Peer-to-Peer Networks

- Distributed *key/value* storage, typically hashes for *keys*
- Range queries (PastryStrings, PHT)
- Pattern matching (Cubit, DPMS)
- Similarity queries (Karnstedt et al.)

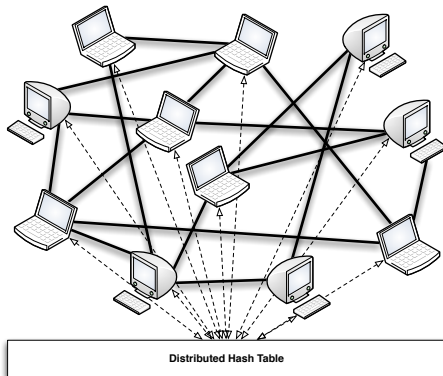


Searching in DHT-based Peer-to-Peer Networks

- Distributed *key/value* storage, typically hashes for *keys*
- Range queries (PastryStrings, PHT)
- Pattern matching (Cubit, DPMS)
- Similarity queries (Karnstedt et al.)
- Our approach: regular expressions

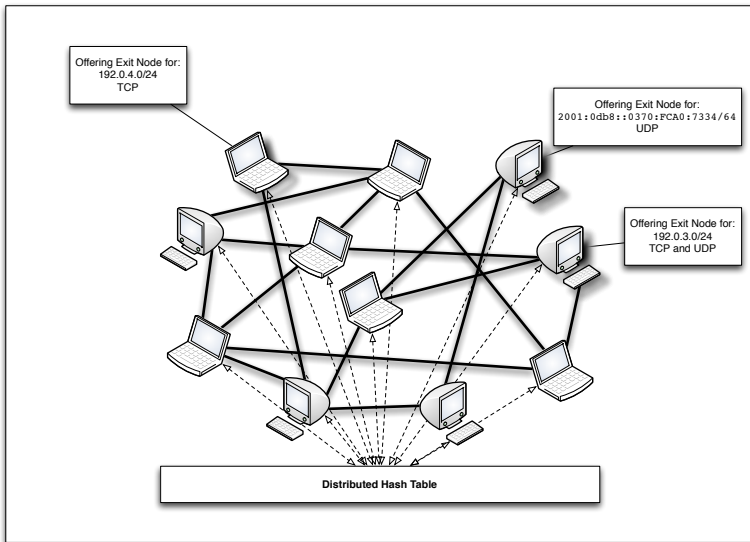


Capability Discovery in Peer-to-Peer Networks



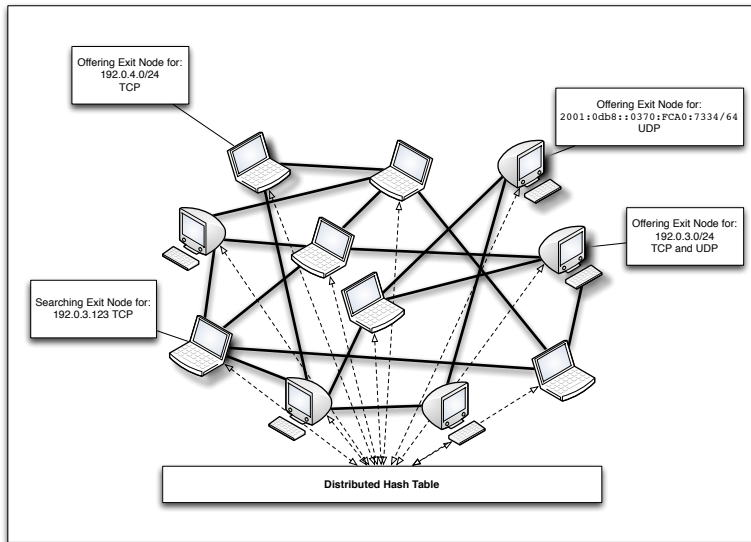


Capability Discovery in Peer-to-Peer Networks



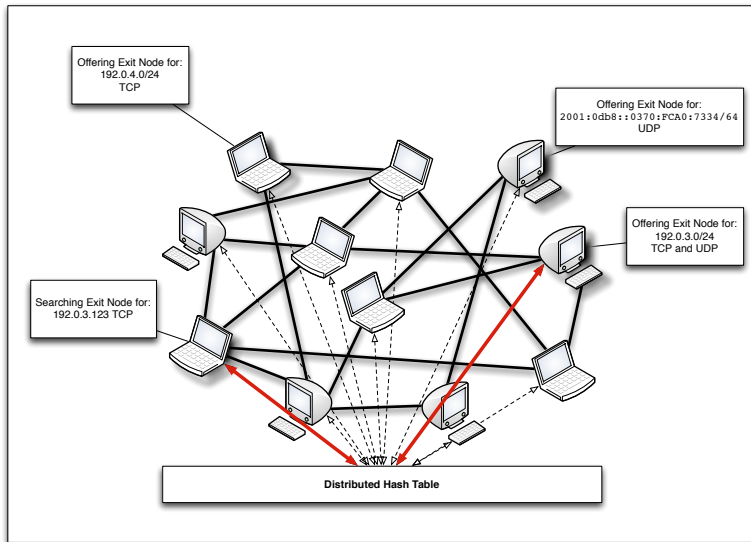


Capability Discovery in Peer-to-Peer Networks



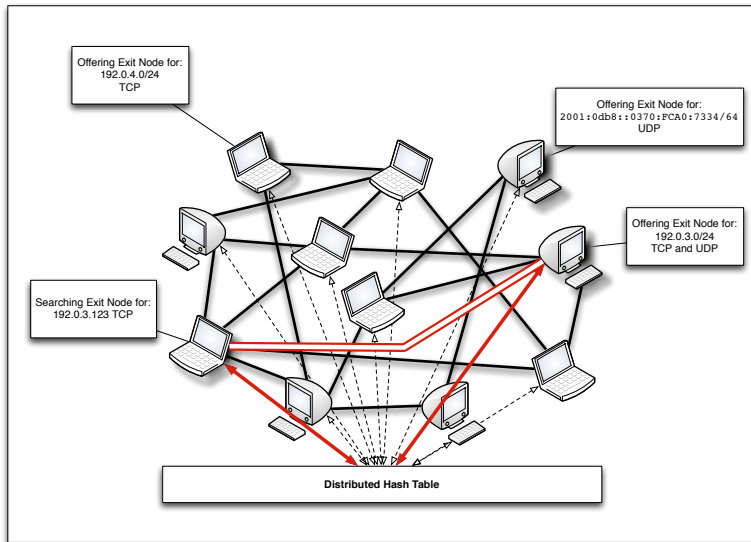


Capability Discovery in Peer-to-Peer Networks





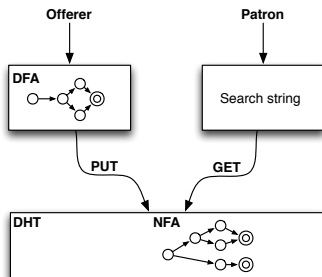
Capability Discovery in Peer-to-Peer Networks





Approach: Idea

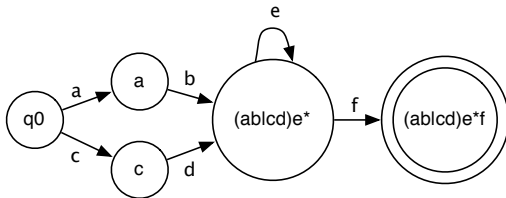
- 1 Offerer creates regular expression describing service
- 2 Regular expression is converted to a DFA
- 3 DFA is stored in the DHT
- 4 Patron matches using a string





Problem: Mapping of States to Keys

Regular expression $(ab|cd)e^*f$ and corresponding **DFA**

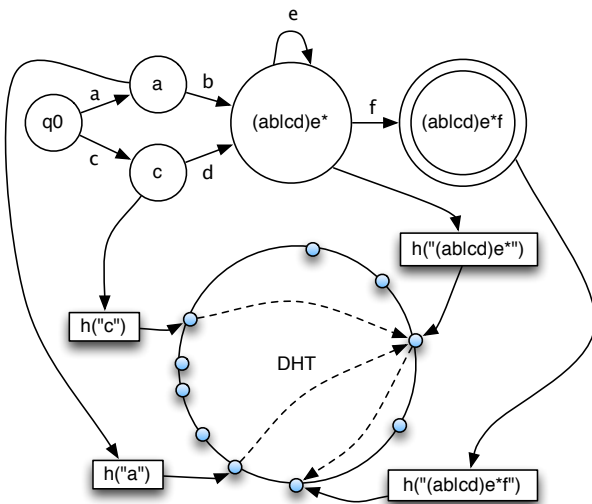


A regular expression is assigned to each state as its identifier.
The hash of the identifier is used as the *key* for DHT PUT.



Problem: Mapping of States to Keys

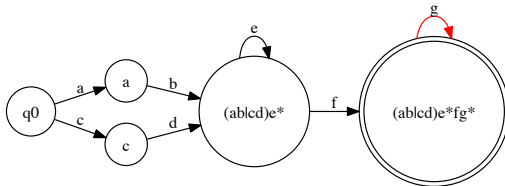
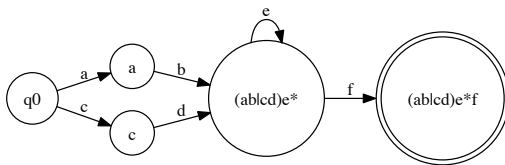
Regular expression $(ab|cd)e^*f$ and corresponding **DFA**





Problem: Merging of DFAs

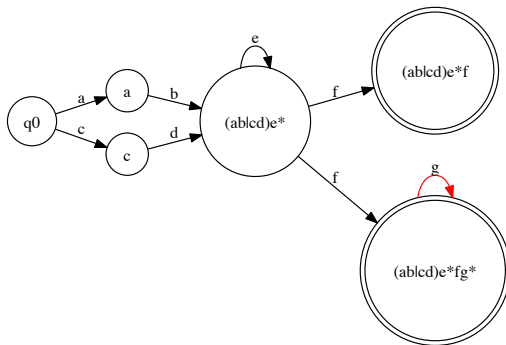
Regular expressions $(ab|cd)e^*f$ and $(ab|cd)e^*fg^*$ with corresponding **DFAs**





Problem: Merging of DFAs

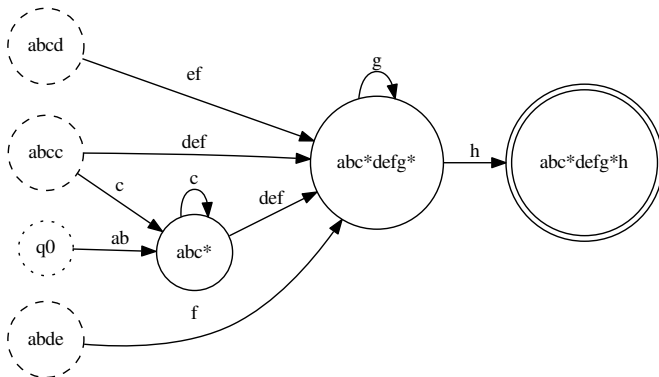
Merged **NFA** for regular expressions $(ab|cd)e^*fg^*$ and $(ab|cd)e^*f$





Problem: Decentralizing the Start State

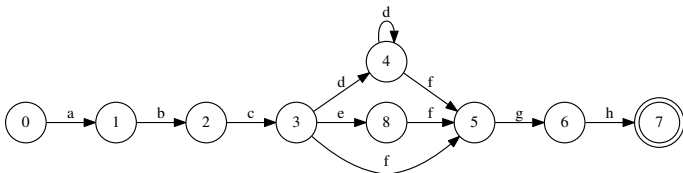
Regular expression: abc^*defg^*h and $k = 4$.





Problem: Optimization (Path compression)

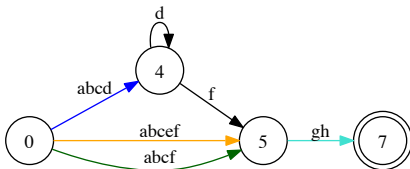
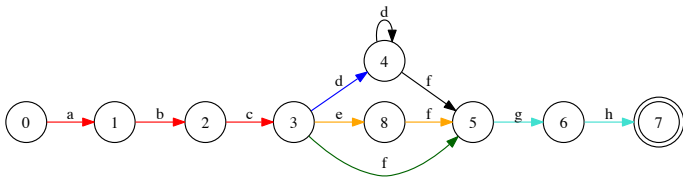
Compressing linear paths in the DFA. Example for $abc(d^*|e)fgh$.





Problem: Optimization (Path compression)

Compressing linear paths in the DFA. Example for $abc(d^*|e)fgh$.



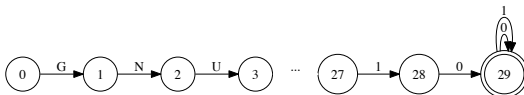


Merging of DFAs with path compression

GNUNET – *VPN*00010000 – *V4TCP*110000000000000000000010(0|1)*

GNUNET – *VPN*00010000 – *V4TCP*1100000000000000000000011(0|1)*

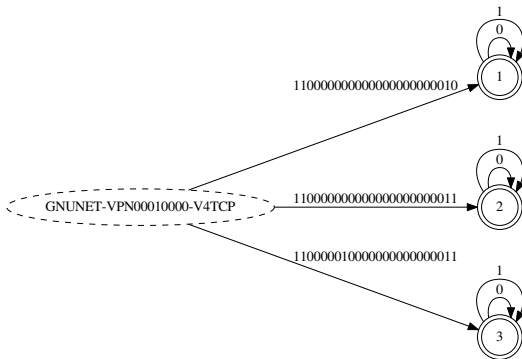
GNUNET – *VPN*00010000 – *V4TCP*1100000100000000000000011(0|1)*





Problem: Path compression length

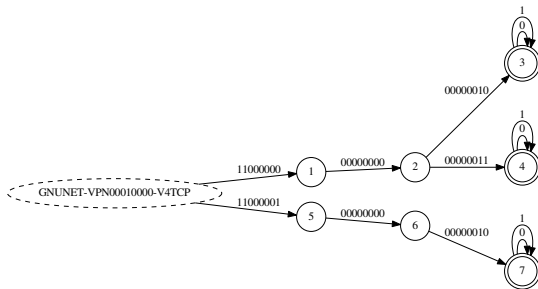
Merging of DFAs with maximal path compression





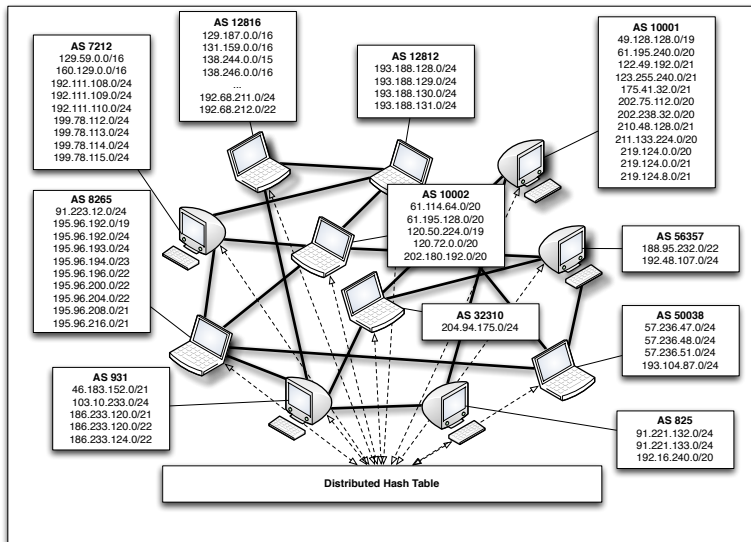
Problem: Path compression length

Merging of DFAs with limited path compression length



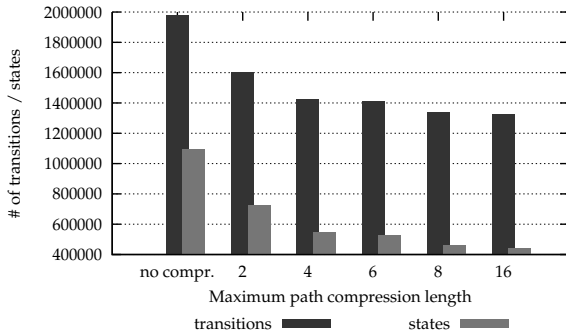


- Implementation in GUNet
- Profiling of Internet-scale routing using regular expressions to describe AS address ranges
- CAIDA AS data set: Real AS data





Number of transitions and states in the merged NFA

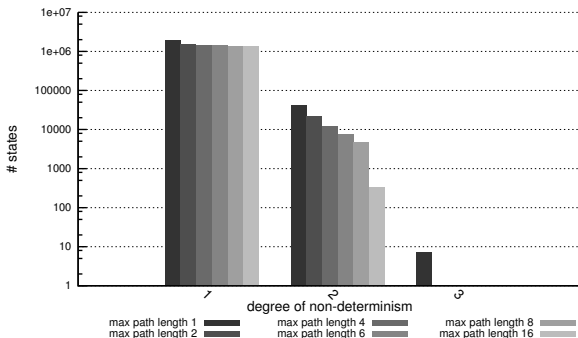


Dataset: All 40,696 ASs



Evaluation: Results of Simulation (2)

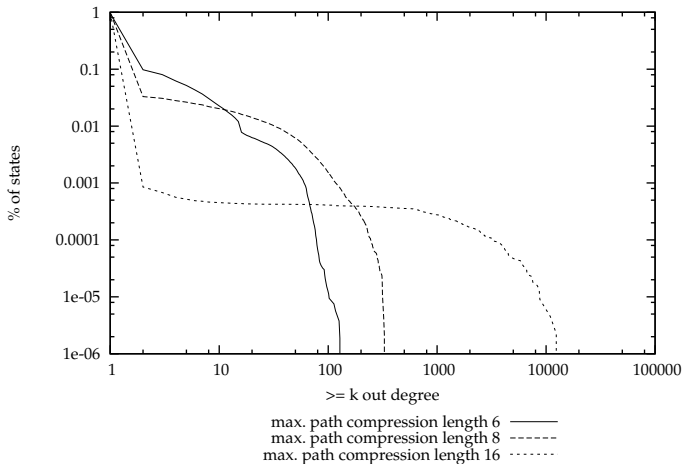
Degree of non-determinism at states in the merged NFA



Dataset: All 40,696 ASs



Evaluation: Results of Simulation (3)

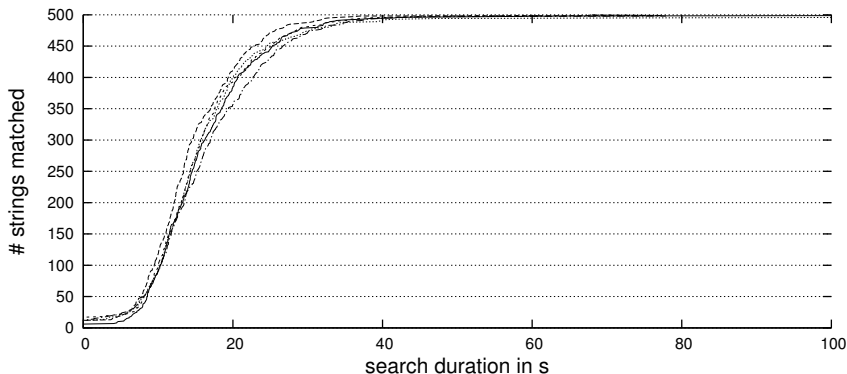


Dataset: All 40,696 ASs



Evaluation: Results of Emulation (1)

Search duration for five runs with 500 randomly connected peers, 500 regular expressions and 500 search strings.





Achievements

- Capability discovery in DHT-based P2P networks using regular expressions
- Linear latency in the length of the search string
- Suitable for applications that can tolerate moderate latency

Future Work

- Use regular expression search in new applications
- Open problem: searching using a regular expression
- Ultra-large scale profiling (SuperMUC)



Thank you!



Appendix



Emulation with subset of the CAIDA AS data

Average per peer	100 peers	500 peers	1, 000 peers
PUT messages	561 kB	1, 231 kB	1, 166 kB
GET messages	291 kB	364 kB	430 kB
RESULT messages	20, 657 kB	762, 485 kB	397, 840 kB
Number of states visited	9	10	9
Path length to accept state	5	4	5
search duration	15 s	30 s	46 s



Example regular expression for AS 14259

```
GNVPN-0001-PAD(000000000111010011000002(0|1)*|10011100011000010010(0|1)*|10101000111001111001001(0|1)*
|10111110000010000100(0|1)*|101111100000100001000010(0|1)*|101111100000100001000011(0|1)*|101111100000100001000100(0|1)*
|101111100000100001000101(0|1)*|101111100000100001000111(0|1)*|101111100000100001001000(0|1)*|101111100000100001001001(0|1)*
|101111100000100001001010(0|1)*|101111100000100001001011(0|1)*|101111100000100001001100(0|1)*|101111100000100001001101(0|1)*
|101111100000100001001110(0|1)*|10111110000010000101(0|1)*|10111110000010000101000(0|1)*|101111100000100001010001(0|1)*
|101111100000100001010011(0|1)*|101111100000100001010100(0|1)*|101111100000100001010101(0|1)*|101111100000100001010110(0|1)*
|101111100000100001010111(0|1)*|101111100000100001011000(0|1)*|101111100000100001011001(0|1)*|101111100000100001011010(0|1)*
|101111100000100001010110(0|1)*|101111100000100001011000(0|1)*|101111100000100001011001(0|1)*|101111100000100001011010(0|1)*
|101111100000100001011011(0|1)*|101111100000100001011100(0|1)*|1011111000001000010111001(0|1)*|1011111000001000010111010(0|1)*
|101111100000100001100111(0|1)*|101111100000100001101000(0|1)*|101111100000100001101001(0|1)*|101111100000100001101010(0|1)*
|101111100000100001101011(0|1)*|101111100000100001101100(0|1)*|101111100000100001101101(0|1)*|101111100000100001101110(0|1)*
|101111100000100001110000(0|1)*|101111100000100001110001(0|1)*|101111100000100001110010(0|1)*|1011111000001000011100110(0|1)*
|101111100000100001110101(0|1)*|101111100000100001110110(0|1)*|1011111000001000011101110(0|1)*|10111110000010000111011101(0|1)*
|101111100000100001111010(0|1)*|101111100000100001111010(0|1)*|1011111000001000011110110(0|1)*|10111110000010000111101110(0|1)*
|10111110000010000111101101(0|1)*|10111110000010000111101110(0|1)*|101111100000100001111011110(0|1)*|1011111000001000011110111101(0|1)*
|10111110000010000111101110(0|1)*|10111110000010000111101110100(0|1)*|10111110000010000111101110111(0|1)*|101111100000100001111011110111(0|1)*
|10111110000010000111101110101(0|1)*|1011111000001000011110111000(0|1)*|10111110000010000111101110001(0|1)*|10111110000010000111101110001000(0|1)*
...
|11011100110(0|1)*|110010011110111011100111(0|1)*|110010011110111011110001(0|1)*|110010011110111011110011(0|1)*
|110010011110111011110101(0|1)*|110010011110111011110110(0|1)*|110010011110111011111011(0|1)*|11001001111011101111100(0|1)*
|11001001111011111111111(0|1)*
```



Example DFA for regular expression AS 14259

